
RNA editing in the pig brain

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REG.NR.: 1588648

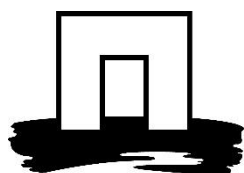
Major thesis Animal Breeding and Genomics (ABG80436)

March 2025

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Examiner: Ole Madsen

Thesis: Animal Breeding and Genomics



WAGENINGEN
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Abstract

Adenosine-to-Inosine (A-to-I) editing, catalysed by the ADAR enzymes, is a post-transcriptional modification prevalent in mammalian tissues and specifically the brain. Although A-to-I RNA editing has been investigated in different mammalian species, including pig, RNA editing at the population level in non-human mammalian species is understudied. Here, we explore the RNA editing landscape in 141 brain transcriptome samples from a population of 48 young female pigs. We identify 15,288 unique RNA editing sites across three brain tissues, namely cortex, hippocampus and hypothalamus. Cortex and hippocampus samples in general showcase a similar RNA editing profile and hypothalamus samples have a more distinct RNA editing signature. The majority of the detected RNA editing sites are located in protein-coding genes and in repetitive elements, such as Short Interspersed Nuclear Elements. Approximately half of the RNA editing sites are found in the 3'UTR region and 2% in the CDS regions. Genes that harbor editing sites in their CDS region code for neuroreceptor proteins or are involved in the regulation of neurotransmission. We observe a correlation between *ADAR2* mRNA expression levels and RNA editing activity in hippocampus samples but not for other tissues or ADAR enzymes. Few sites were differentially edited between cortex and hippocampus and several hundred were differentially edited between hypothalamus and cortex, as well as hypothalamus and hippocampus. To our knowledge, this is the first study that looks into the brain RNA editing landscape in a pig population.

1. Introduction

1.1 Definition of RNA editing and A-to-I substitutions

RNA editing was first used as a term in 1986 (Benne et al., 1986) to describe the insertion of four uridines in the mitochondrial mRNA of the cytochrome oxidase subunit II (*COXII*) gene in a protozoon species. Since then, it has been used to refer to the insertion, deletion and conversion of bases in transcribed RNA molecules with no matching bases in the template DNA. RNA editing is considered part of the epitranscriptome, which refers to the set of all dynamic biochemical alterations occurring on transcribed RNA molecules that regulate gene expression and grant cells adaptive capabilities to environmental changes (Christofi & Zaravinos, 2019). In contrast to other epitranscriptomic modifications, RNA editing modifications can change the ribonucleotide sequence itself.

Although insertions and deletions through the mechanism of RNA editing have not been found in animals, base conversions, however, are indeed widespread (Gray, 2012). The prevalent manifestation of RNA editing in higher eukaryotes is the substitution of Adenosine to Inosine (A-to-I) through deamination of Adenosine (Fig.1). The deamination reaction is catalysed by the Adenosine Deaminase Acting on RNA (ADAR) proteins. A-to-I editing is considered as the canonical form of RNA editing (Li & Church, 2013), and to a lesser extent, the Cytosine to Uracil (C-to-U) substitution (Blanc & Davidson, 2003) is present as well. Inosine pairs with Cytosine and is therefore interpreted as a Guanosine by the cellular machinery (Li & Church, 2013). Initially thought to be an uncommon phenomenon, the advance of Next Generation Sequencing technologies have enabled the detection of a plethora of transcriptome-wide RNA editing events across different species (Li et al., 2017).

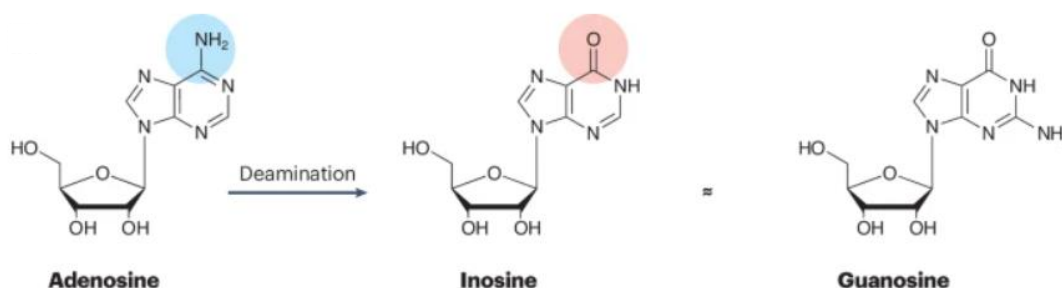


Fig 1. Deamination of Adenosine into Inosine catalysed by ADAR, taken from (Li & Walkley, 2025).

1.2 The ADAR enzymes and their role in RNA editing

The ADAR enzymes typically target internal double stranded RNA, the latter being capable of forming structures that resemble viral RNA. MDA5 is a cytoplasmic protein that binds to long double stranded RNA and triggers the innate immune response by forming filaments with double stranded RNA, which activate the type I interferon response through downstream signalling (Li & Walkley, 2025). Transcribed RNA molecules, including premature mRNA, can form double stranded formations due to the presence of repeat elements in their sequence. Editing of self RNAs by ADAR1 is therefore thought to prevent triggering of the innate immune response, since edited RNAs possess an altered secondary or tertiary structure that prevents MDA5 from forming filament structures that signal the immune system (Liddicoat et al., 2015; Mannion et al., 2014). The activity of ADAR1 also has an effect on genome evolution (Eisenberg & Levanon, 2018): The insertion of new mobile repetitive elements in genomic regions could cause an innate immune response during their transcription and double strand formation. However, the editing of these transcripts leads organisms to tolerate the addition of new mobile genetic elements and enhance genome plasticity (Eisenberg & Levanon, 2018).

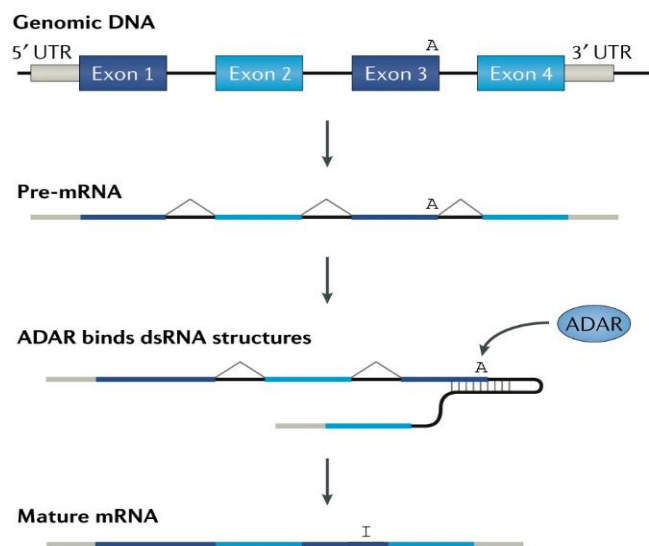


Fig 2. Pre-mRNA molecules can take a double-stranded formation that is then targeted by ADAR which catalyses the A-to-I conversion. If the sites lie within an exon, the Inosine will be present in the mature mRNA. Taken from (Eisenberg & Levanon, 2018).

The enzymes that catalyse A-to-I editing by the deamination of Adenosine are the ADAR proteins, namely the ADAR1 and ADAR2 enzymes, coded by *ADAR* and *ADARB1*, respectively (Nishikura, 2016). ADAR1 is commonly expressed among different cell types and tissues in mammals, however ADAR2 expression is more pronounced in the central nervous system and brain with lower expression values compared to ADAR1 (Nishikura, 2016). Two ADAR1 isoforms are known to exist, namely p110 which is primarily located in the nucleus and p150 which can be found both in the cytoplasm and nucleus, while ADAR2 is mostly constrained in the nucleus (Samuel, 2011). That being said, the bulk of premature mRNA editing occurs in the

nucleus (Samuel, 2011). Furthermore, ADAR3, coded by *Adarb2* is strictly expressed in the brain, can translocate to both nucleus and cytoplasm and is thought to lack the deaminase catalytic activity, likely acting as an antagonist to the other two ADARs (Deffit & Hundley, 2016).

1.3 Distribution of RNA editing sites in genomic regions and repetitive elements

A-to-I editing sites are present throughout the genome but are concentrated in non-coding sequences of genes. These are mostly introns and 3' UTR regions (Nishikura, 2010) that are enriched with repetitive retrotransposon elements, such as the Alu elements in the human genome (Sela et al., 2007) and Porcine Repetitive Elements in the pig genome (Chen et al., 2021). RNA editing can also occur in the coding sequences of genes but in significantly lower numbers compared to intron and 3'UTR sites (Rosenthal & Seeburg, 2012). It was previously thought that ADAR1 predominantly edits sites in repetitive elements and ADAR2 edits recoding sites and sites in non-repetitive regions (Tan et al., 2017), recent studies on mouse genetic models manipulating the expression of the ADAR enzymes, however, pinpoint that sites in repetitive elements are also edited by ADAR2 (Heraud-Farlow & Walkley, 2020).

A-to-I editing can take place at individual adenosines in a selective fashion or at multiple adenosines that are close to one another in the sequence, with the latter called hyper-editing (Wahlstedt & Ohman, 2011). In the human genome, close to 1.6 million editing sites have been detected, the majority of which are in *Alu* repetitive elements (Bazak et al., 2014) and similarly to *Alu* elements in primates, PRE-1 Short Interspersed Nuclear Elements (SINEs), which constitute the major SINE element in the porcine genome, also accumulate the largest share of editing sites (Funkhouser et al., 2017; Huang et al., 2021). Importantly, both edited and un-edited transcripts can co-exist in a cell at a given time, with varying proportions of edited to un-edited transcripts (alternatively the editing rate or editing levels) depending on the tissue and developmental stage of the organism (Lo Giudice et al., 2020). In general, RNA edited sites located in repetitive elements have a low editing rate of less than 5%) and sites in coding sequences tend to have higher editing rates (Roth et al., 2019).

1.4 Functional effects of RNA editing sites are context-dependent

The biological implications of RNA editing depend on the genomic region that the editing sites are located in. Editing that takes place in the 3'UTR region of a transcript can affect the binding site of microRNA molecules and influence the suppression levels of that transcript, regardless of the fact that microRNAs themselves are also edited, leading to changes in their binding capacity and specificity (Nishikura, 2016). Moreover, editing that occurs near splice junctions can exclude or include intronic and exonic elements, thereby regulating the expression of different protein isoforms through altering the splicing pattern (Hogg et al., 2011). The folding of transcribed RNA molecules and consequently their structural stability has also been shown to be impacted by RNA editing (Wang et al., 2013). Nonetheless, the majority of editing sites lying in repetitive elements are believed to be of low functional importance (Heraud-Farlow & Walkley, 2020).

Coding sequence editing leading to a non-synonymous mutation is referred to as a recoding site. Although recoding sites are scarce compared to non-coding sites, they can diversify the encoded isoforms that genes produce. One of those isoforms are produced by the *IGFBP7* gene that codes for an insulin-like growth factor protein (Larsen et al., 2016b). More importantly, however, recoding sites are concentrated in transcripts that are expressed in neural tissues and expand the functional capabilities of key neural genes (Behm & Öhman, 2016). RNA editing has been shown to regulate neurotransmission across mammalian species, with the α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) glutamate receptor being the most well-studied example. The substitution of glutamine to arginine (Q/R) in the GluA2 subunit of the AMPA glutamate receptor (Fig.3) through recoding is vital in ensuring Ca^{2+} impermeability to glutamate receptors (Lomeli et al., 1994; Seeburg & Hartner, 2003). Early postnatal death is observed in mice that are experimentally deficient of *Adar2*, possibly due to excessive Ca^{2+} influx (Higuchi et al., 2000). Editing in the brain can also impact signal transduction, as in the case of the serotonin receptor HTR2C in mammals, including pig (Larsen et al., 2016a). Specifically, five nucleotides in exon 5 of the HTR2C receptor can be edited to give rise to different isoforms, all of which have varying effects on the interaction between the receptor and its affiliated G protein, hence modulating the serotonin signal pathway. Finally, editing influences the amount of cell membrane receptors in mammals, as is evident for AMPA, GABA receptors and K^+ ion channels (Behm & Öhman, 2016; Huang et al., 2021) and is known to modulate the activity of the brain-specific Nova1 splicing factor (Irimia et al., 2012). Interestingly, mouse genetics studies have shown that, excluding the case of the AMPA receptor subunit, removal of all edited protein isoforms is tolerated by organisms (Heraud-Farlow & Walkley, 2020).

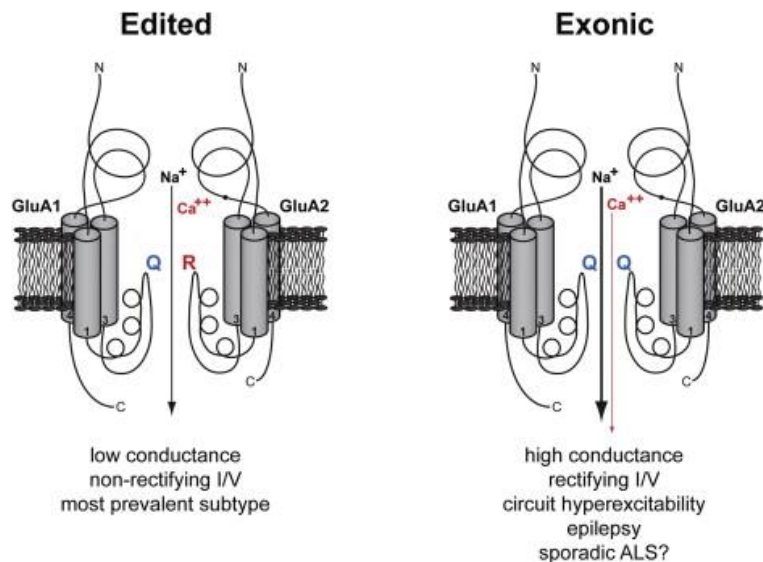


Fig 3. The two out of four subunits of edited (left) and un-edited (exonic, right) versions of the AMPA receptor are illustrated. The edited version with the Q/R substitution has a low conductance for Ca^{2+} and is the most prevalent version. The un-edited version has a high conductance for Ca^{2+} and is lethal post-natal. Taken from (Rosenthal & Seeburg, 2012).

1.5 RNA editing in disease and the pig as a biomedical model

Aberrant ADAR expression has been linked with several diseases. Mutations in the *ADAR1* gene are one of the causes of Aicardi-Goutières syndrome (Rice et al., 2012), which is a disease characterized by encephalopathy and brain atrophy, while a variety of different mutations have been found in patients with Dyschromatosis symmetrica hereditaria, a skin pigmentation disorder (Kono & Akiyama, 2019), as well as dystonia (Herbert, 2020). Mutations of the *ADARB1* gene, encoding ADAR2, have been implicated in seizures, microcephaly and intellectual disability (Tan et al., 2020). Dysregulated RNA editing landscapes in the human brain have been associated with autism (Tran et al., 2019), epilepsy (Srivastava et al., 2017), glioblastomas (Silvestris et al., 2019), Alzheimer's disease (Ma et al., 2021) and schizophrenia (Breen et al., 2019).

Humans and pigs shared a common ancestor roughly 79-97 million years ago (Kumar & Hedges, 1998). The pig has been utilized extensively to model specific human diseases due to its similar physiology and anatomy with human and feasibility to produce knockouts and transgenics (Vodička et al., 2005). Regarding the study of the brain and neurodegenerative disorders, pig as a disease model is superior to that of mouse, since pigs possess a gyrencephalic brain morphology and a white to grey matter composition that are similar to humans, have quite a larger brain when compared to mice and are more accessible compared to nonhuman primate brains (Lunney et al., 2021).

1.6 Context and goals of the present study

In the present, literature regarding RNA editing in the pig brain is relatively limited. Studies have examined RNA editing activity across different tissue types in pigs, some of them including the prefrontal cortex which is a brain subregion (Adetula et al., 2021; Wang et al., 2019; Zhang et al., 2019). A study focused on multiple pig brain subregions (Huang et al., 2021) from four pigs is the most extensive and high-throughput study currently in the field. A commonality in literature is that, unlike studies in humans (Cuddleston et al., 2022; Silvestris et al., 2019), only a small number of animals are sampled. Consequently, research on RNA editing in the pig brain at the population level is currently missing in the field.

The main goals of the present study are to investigate the brain RNA editing landscape in a population of gilts and compare RNA editing activity across three different brain subregions, namely prefrontal cortex, hippocampus and hypothalamus. The amount of shared and unique RNA editing sites, along with the RNA editing activity at the sample level, were examined between the three tissues. We studied the distribution of RNA editing sites in both repetitive elements and genomic coding regions with a particular focus on RNA editing sites in genes. Lastly, we aimed to study differential RNA editing between tissues, as well as a possible association between ADAR mRNA expression levels and RNA editing activity of samples per tissue.

2. Methods

2.1 Experimental design and RNA extraction protocol

The dataset utilized in the present study was obtained from an experiment conducted as described in (Luo et al., 2022). A population of 48 gilts were randomly selected from 384 three-way crossbreds (Synthetic * (Large White * Landrace)). With the completion of weaning at a mean age of 4 weeks, the piglets were allocated to a 2 x 2 factorial design to examine the effects of allostatic load and environmental conditions, each condition having 12 piglets in total. At 8 to 9 weeks old the gilts were sacrificed in a single batch and tissue samples from the prefrontal cortex, hippocampus and hypothalamus were retrieved from each gilt and preserved in RNAlater at -80°C, where hypothalamus samples from 3 pigs were omitted due to failure during the sampling procedure. All animals were genotyped on the Porcine 660K SNPchip and imputed to sequence using Beagle software v5.4 (Browning et al., 2018).

The Qiagen RNeasy kit was used to extract total RNA based on the manufacturer's protocol, followed by quality control of the extracted RNA, quantification with Thermo Fisher's NanoDrop 1000 and Invitrogen's Qubit RNA BR Assay kit and RNA integrity evaluation with Agilent Technologies Bioanalyzer Agilent RNA 6000 Nano kit, where all samples had an RNA Integrity value above 9. RNA was poly(A) enriched using NEBNext® Poly(A) mRNA Magnetic Isolation Module (New England Biolabs), followed by library preparation with the NEBNext® Ultra™ Directional RNA Library Prep Kit (New England Biolabs). Sequencing was performed on the Illumina NovaSeq 6000 platform for the generation of stranded paired end reads of 150 bp in length.

2.2 RNA-seq read mapping and preprocessing

We performed quality and adapter trimming on raw RNA-seq reads with Trim Galore 0.3.7 (Krueger, 2015). We then mapped the processed RNA-seq data to the Sscrofa11.1 (Ensembl version 104) reference genome with STAR 2.7.10b (Dobin et al., 2012) after indexing the genome, with an average of 42 million reads per sample. STAR outputted a coordinate-sorted BAM file for each sample and we removed potential PCR duplicates, with an updated average of 29.5 million unique reads per sample. Next, we calculated MD tags for each BAM file with SAMtools 1.22.1 (Li et al., 2009) *markdup* and *calmd* commands, respectively. The processed BAM files were visualized in JBrowse (Buels et al., 2016).

2.3 *De novo* identification of RNA editing sites and filtering

We conducted *de novo* RNA editing identification on the BAM files separately based on tissue with JACUSA2 v2.1.16 (Piechotta et al., 2022) using the *call-1* method and the mode for first strand-specific sequences. To ensure selection of true RNA editing sites and artefact removal, a set of strict filters was utilized as follows: (i) reads needed to have a mapping quality score of at least 20, (ii) candidate editing sites required a PHRED quality score of at least 24, (iii) not lie in the first and last 6 bases of a read, (iv) lie at least 6 bases away from a splice site, (v) lie at least 10 bases away from an INDEL position and (vi) must not lie in homopolymeric regions larger than 7 residues long. Also, candidate editing sites which mapped to the set of 33,739,170 SNP positions that were imputed to Whole Genome Sequencing data from the 660K SNP genotyping dataset of the pig population were excluded, because these represent true genomic variants segregating in the population. At the sample level, RNA editing sites with more than 2 alleles present were also filtered out.

We further filtered the variant calling output from JACUSA2 with the auxiliary *JACUSA2helper* (Piechotta et al., 2022) package to identify RNA editing sites. Putative canonical RNA editing sites were selected by filtering for sites with an A reference and a G variant. To quantify the level of RNA editing that occurs in a given candidate editing site, we divided the number of edited reads covering that site by the total coverage for that site, or in other words, the editing rate. An A-to-G site was deemed a true editing site if at least 5 samples in a tissue type had (i) coverage of at least 10 reads, (ii) at least 2 edited reads, and (iii) an editing rate of at least 0.05 for that given site. All the RNA editing events across samples for that given site would then be included in the analysis, provided that they had a coverage of at least 10 reads, since these events would more likely represent an editing signal rather than technical or biological noise. To remove any further genomic variants that were missed from the imputed sequence data, we discarded candidate sites with a mean larger than 0.95 for samples with a non-zero editing rate, with the exception of the (Q/R) site that lies in the *GRIA2* gene, which is the only known site to be edited almost 100% of the time in the mammalian brain (Lomeli et al., 1994).

2.4 Tissue comparisons and sample RNA editing metrics

To perform Principal Component Analysis (PCA) on the samples, we structured the editing rates of all detected RNA editing sites across samples into a matrix and conducted PCA using R's built-in *prcomp* function.

To compare the number of unique and shared editing sites between tissues, we constructed a Venn diagram of unique RNA editing sites per tissue with the *VennDiagram* (Chen & Boutros, 2011) package. To measure per-sample RNA editing activity, the Overall Editing Level (OEL) was calculated as the aggregated number of edited reads of all editing sites in a single sample divided by the aggregated total reads at those sites. We used the non-parametric Friedman's test, which is similar to ANOVA

but for related groups, to assess tissue effect on the number of sites and OELs, followed by pairwise Wilcoxon signed-rank tests.

2.5 Annotation of detected RNA editing sites

For each tissue, we converted the final list of RNA editing sites into BED format and used bedtools v2.31.0 (Quinlan & Hall, 2010) *intersect* to annotate the genomic regions in which the editing sites were located by intersecting them with the genome annotation GTF file. We also intersected the sites with RepeatMasker annotations (Chen, 2004) to identify those occurring in repetitive elements. The RepeatMasker annotation file was downloaded from the UCSC Table Browser.

Next, we converted the editing site lists to VCF format to predict the mutation consequences of recoding sites using Ensembl's Variant Effect Predictor (VEP) (McLaren et al., 2016). Finally, we performed Gene Ontology (GO) enrichment analysis on genes harboring at least one editing site in their coding sequence using ShinyGO v0.85 (Ge et al., 2019), providing Ensembl Gene IDs as both foreground and background gene sets. GO terms were considered significant at a False Discovery Rate (FDR) threshold of 0.05.

2.6 ADAR expression levels and association with RNA editing activity

To examine whether tissue differences in RNA editing can be explained by differences in ADAR expression levels, we extracted the gene mRNA expression levels of the ADAR1, ADAR2 and ADAR3 enzymes, measured in Transcripts per Million, from a previously conducted gene expression study on the same dataset. For each tissue type, we constructed a simple linear model by regressing the overall editing level of samples against sample mRNA expression levels of each ADAR enzyme, for a total of (3x3) 9 models and estimated the correlations with Pearson's *r*. We accounted for multiple testing by correcting p-values with the Benjamini Hochberg (BH) procedure to control for the FDR at a value of 0.05 (Benjamini & Hochberg, 1995).

2.7 Differential editing between experimental conditions and tissues

Differential editing was conducted between experimental factors and between tissues. We tested for differential editing between tissues with the non-parametric two-tailed Wilcoxon signed-rank test for paired data using the base R *wilcox.test* function with the *paired* parameter set to TRUE. One test was conducted for all pairwise comparison between tissues, for a total of three tests. For a given editing site to be included in the differential analysis, at least 12 samples on each tissue had to be edited (Editing Rate >0) on that particular site. No threshold for the effect size (median difference) was chosen. We applied multiple testing correction after each test with the

Benjamini Hochberg (BH) procedure to control for the FDR at a value of 0.05 (Benjamini & Hochberg, 1995).

We investigated differential editing between experimental factors with a non-parametric and parametric approach. All tests conducted were examining one tissue at a time. For the non-parametric approach, we used the two-tailed Mann-Whitney test with base R *wilcox.test* function and the *paired* parameter set to FALSE, also followed by multiple testing correction with BH for controlling the FDR at 0.05. For the parametric approach, we constructed three different linear models, one model looking at the main effect of one experimental factor at a time, one model with both experimental factors and one model with both experimental factors including their interaction. Linear models were constructed with the base R *glm* function.

2.10 Plot visualizations

We extracted the flanking sequences of identified editing sites and random control Adenines with the help of bedtools *flank* and *getfasta* functions. Flanking sequence logos plots were then made with the *ggseqlogo* (Wagih, 2017) R package. Excluding the Venn diagram and flanking sequence logo plot, all other plot visualizations were done with *ggplot2* (Wickham, 2011).

3. Results

3.1 Filtering of candidate RNA editing sites

RNA-seq reads mapped to the Sscrofa11.1 reference genome at an average of 91% per sample. After deduplication of mapped reads, an average of 29 million unique reads per sample were used for de novo identification of RNA editing sites. Each guilt in the population of 48 pigs was sampled from three different brain-subregions, namely cortex, hippocampus and hypothalamus for a total of 141 samples excluding three hypothalamus samples. The number of initial mismatches with Adenine as a reference base were 4,647,769 for cortex, 4,906,189 for hippocampus and 3,805,975 for hypothalamus (Table 1), aggregated across all the samples for each tissue type. By filtering only for Adenine/Guanine mismatches to detect canonical A-to-I RNA editing, the number of candidate sites was reduced by about half for each tissue. This was followed by the application of strict filters to distinguish between true editing signal and technical noise (see: Methods, *De novo* identification of RNA editing sites and filtering), which had more than a ten-fold reduction in the number of candidate editing sites for all three tissues. Selecting only for sites that passed these filters in at least 5 samples per tissue resulted in another ten-fold reduction in the number of detected sites. Finally, the additional filter implemented for editing sites with an editing rate close to 1, that were more likely genomic variants in the pig population, removed about 50 genomic variants on each tissue.

	Cortex	Hippocampus	Hypothalamus
Sites with A as reference	4,647,769	4,906,189	3,805,975
Sites with A as reference and G as variant	2,488,173	2,652,071	1,999,674
Sites with editing rate ≥ 0.05 , number of edited reads ≥ 2 , coverage ≥ 10	136,038	144,954	120,181
Sites where above filters hold for at least 5 samples	9,691	11,206	8,946
Final sites after additional genomic variant filtering	9,636	11,155	8,902

Table 1. Number of surviving sites after application of filters during de novo identification.

3.2 Principal Component Analysis of samples

To verify the validity of the analysis, we explored whether sample stratification by tissue is possible through the RNA editing profiles of samples. Principal Component Analysis (PCA) was performed on the detected RNA editing sites across all cortex, hippocampus and hypothalamus samples. The PCA (Fig.4) showed a clear clustering of samples based on tissue. Cortex and hippocampus appeared more similar to one another, as opposed to hypothalamus which was more distinct than the other tissues. A highly similar clustering of tissues is evident in the PCA on gene expression levels of the same samples (data not shown). A closer pairing of cortex with hippocampus and a segregation of the hypothalamus has been previously shown in a brain-focused transcriptomics analysis of human, pig and mouse species (Sjöstedt et al., 2020), which can be explained by the higher cellular heterogeneity of the hypothalamus.

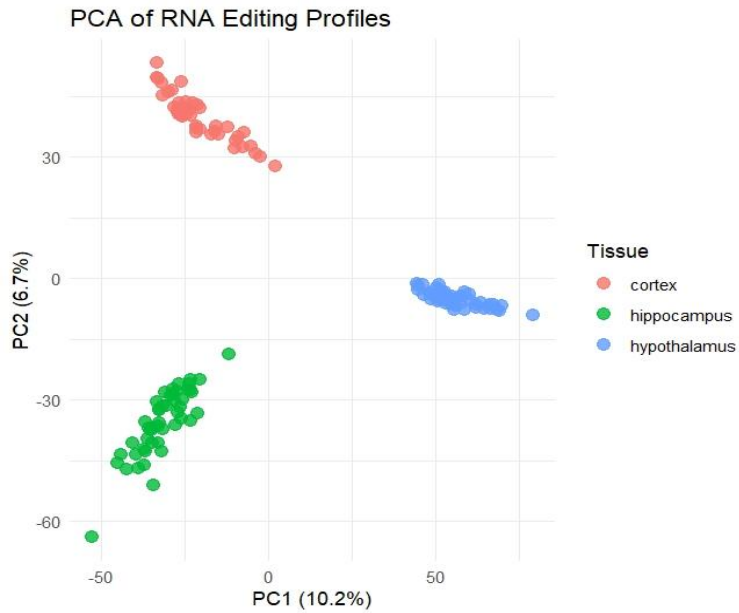


Figure 4. *Principal Component Analysis plot of RNA Editing profiles of all samples.*

3.3 RNA editing across tissues: unique & shared RNA editing sites

The analysis yielded 15,288 unique RNA editing sites in all three tissues in the pig genome. Of these, 7,956 were unique sites from the positive strand and 7,332 were from the negative strand. Out of the 15,288 unique editing sites that were identified (Fig.5), 61% were common between at least two tissues, while nearly one third were present in all three tissues. Cortex and hippocampus shared the most editing sites out of all tissue pairs at 7,466 sites. Hippocampus exhibited the largest amount of identified editing sites at 11,155, followed by cortex at 9,636 and hypothalamus at 8,902.

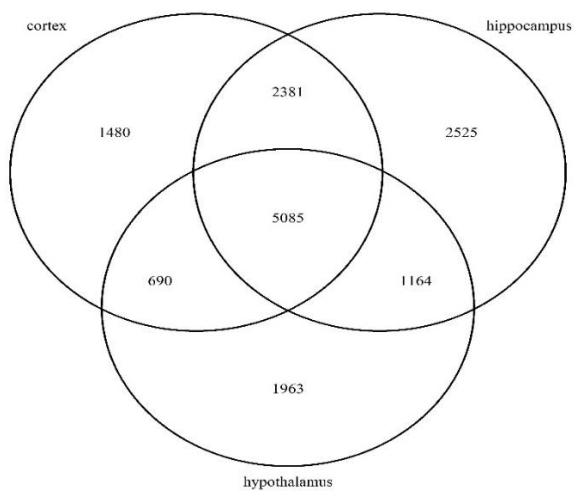


Figure 5. *Unique and shared editing sites across cortex, hippocampus & hypothalamus.*

3.4 RNA editing activity at the sample level

RNA editing activity at the sample level was measured with two different metrics. The first was the total number of editing sites in a sample and the second was the overall editing level, which is the sum of edited reads divided by the sum of reads found in all editing sites in a sample. Friedman's test showed that both number of editing sites ($p < 0.001$) and overall editing levels ($p < 0.001$) of samples were significantly different among tissue types. The median number of editing sites was significantly higher in the hippocampus compared to both the cortex ($p < 0.001$) and the hypothalamus ($p < 0.001$; Fig. 6a). Additionally, the cortex showed a significantly greater number of sites than the hypothalamus ($p = 0.004$).

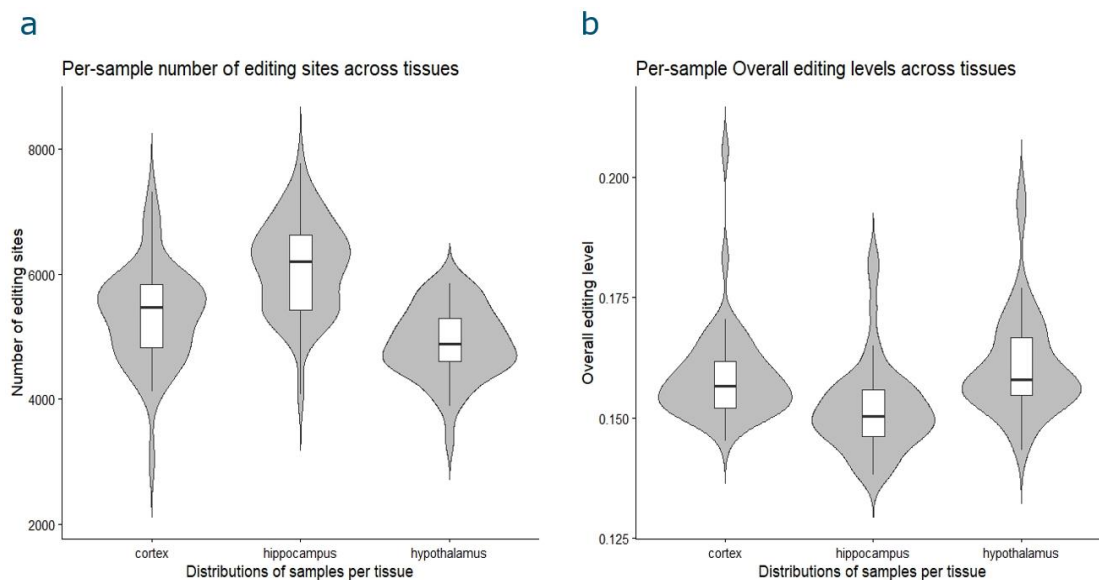


Figure 6. a) *Distribution of number of editing sites per sample across tissue types.* b) *Distribution of overall editing levels per sample across tissue types.*

We argue that the detection of editing sites proportionally depends on the amount of sequencing data available (Tran et al., 2019). Normalizing the number of editing sites of each sample by the total unique mapped reads for that sample yielded a somewhat different picture. The median number of sites per million unique reads for hippocampus samples was significantly higher compared to cortex ($p < 0.001$) samples, but weakly significantly higher than hypothalamus ($p = 0.012$). The median number of sites per million unique reads for hypothalamus samples was also weakly significantly higher than that of cortex. Therefore, the change in how the tissue types compare to one another, that occurs when correcting for sequencing depth, showcases the coverage bias of this metric (Roth et al., 2019).

The overall editing levels, on the other hand, highlight a different aspect of the RNA editing activity in the samples (Fig. 6b). The hippocampus median overall editing level was significantly lower than both cortex ($p < 0.001$) and hypothalamus ($p < 0.001$), while hypothalamus and cortex medians were not found to be significantly different. Taken together with the number of editing sites, we conclude that although hippocampus samples have the largest number of editing sites when compared to

cortex and hypothalamus, the editing sites of hippocampus samples are lowly edited when compared to the other two tissues. It should also be noted that the number of editing sites as a metric is biased towards sites in lowly expressed genes, while the overall editing level is biased towards sites in highly expressed genes.

3.5 Analysis of RNA editing sites across genomic regions and repetitive elements

About 80% of unique editing sites were part of protein-coding genes and the rest were in intergenic regions (Fig.7a). Roughly 50% of all sites were found in 3'UTR regions, a noticeable 30% were found in introns and a mere 2.2% and 0.7% in CDS and 5'UTR regions, respectively. These findings differ from what is commonly reported in literature (Huang et al., 2021; Levanon et al., 2004), where intronic regions are the genomic features with the largest share of editing sites in pig and human. Since the RNA extraction step relied on enrichment for polyadenylated mRNA, we anticipated that more sites in 3'UTR regions would be captured, compared to sites in introns. Consistent with previous observations, the bulk of editing sites are lowly edited (Fig.7b).

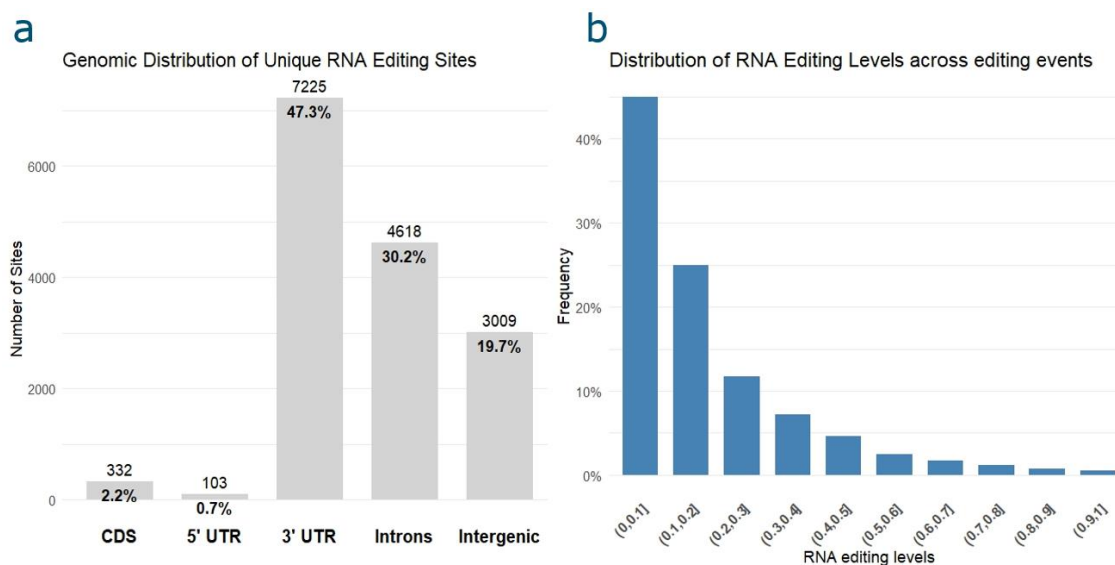


Figure 7. a) *Distribution of total unique canonical RNA editing sites in features in the pig genome across tissues.* b) *Distribution of RNA editing levels of all detected editing events.*

Regarding the distribution of editing sites in repetitive elements, approximately 9 out of 10 RNA editing sites (Fig.8a) are harbored within a repetitive element. Specifically, the vast majority (86.8%) of sites were lying in a SINE element, with just 2.8% of sites in LINES and 2.5% in various elements like LTR or simple repeats. The seven most common repetitive elements (Fig.8b) all belong to the PRE-1 (Porcine Repetitive Element) family (Fang et al., 2012), with Pre0_SS being the prominent PRE-1 element, as is seen in other studies as well (Funkhouser et al., 2017; Huang et al., 2021).

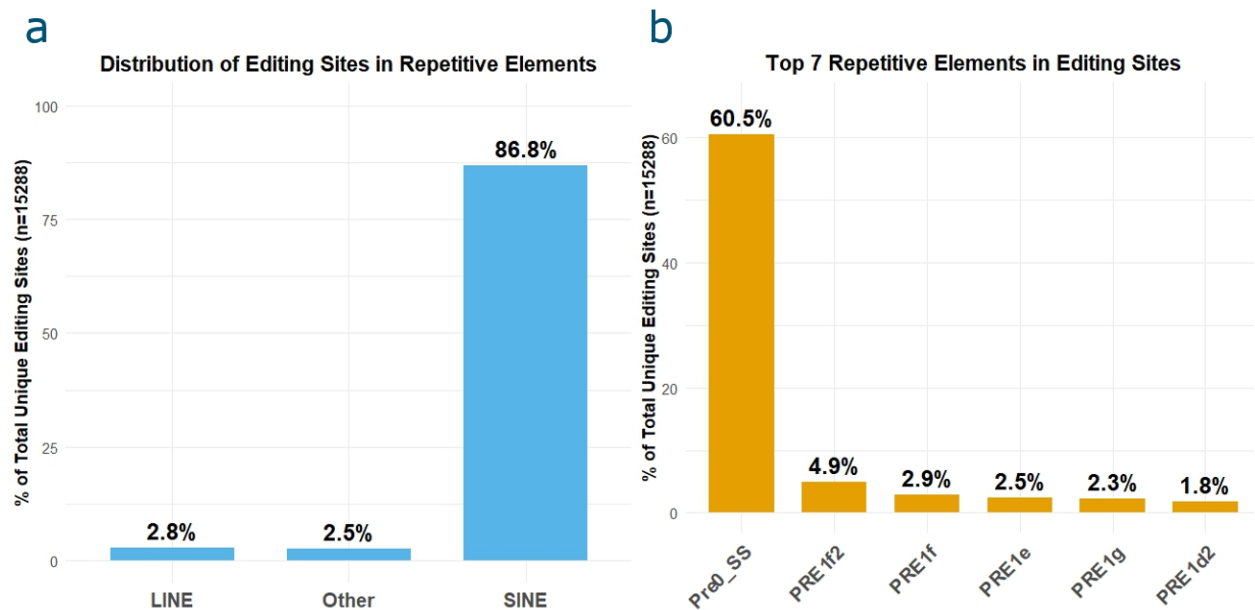


Figure 8. a) *Distribution of total unique canonical RNA editing sites in repetitive elements.* b) *The seven most common repetitive elements that editing sites are found in.*

3.6 RNA editing sites in CDS regions and associated genes

Although editing sites in CDS regions are quite few when compared to other genomic regions, they can have a stronger functional impact. Across the three tissues, 332 unique RNA editing sites were detected in the CDS region of 167 genes. Out of these sites, 207 lead to a non-synonymous mutation, meaning they are recoding sites, and 125 to a synonymous mutation (Fig.9a). The most common amino acid substitutions were Serine to Glycine, Glutamine to Arginine and Arginine to Glycine (Fig.9b). The majority of genes with CDS editing sites had 1 to 3 editing sites, with a few notable exceptions. In particular, the SON gene had 10 recoding sites and 9 additional synonymous editing sites, which codes for the SON RNA-binding protein that functions as a mRNA splicing cofactor of cell-cycle and DNA-repair transcripts. Note that SON protein isoforms are quite large, exceeding 2,000 amino acids, hence the high number of editing sites. Also, the DDX18 gene, with a known transcript 786 amino acids long, had 8 recoding sites and 8 additional synonymous editing sites and codes for an ATP-dependent RNA helicase.

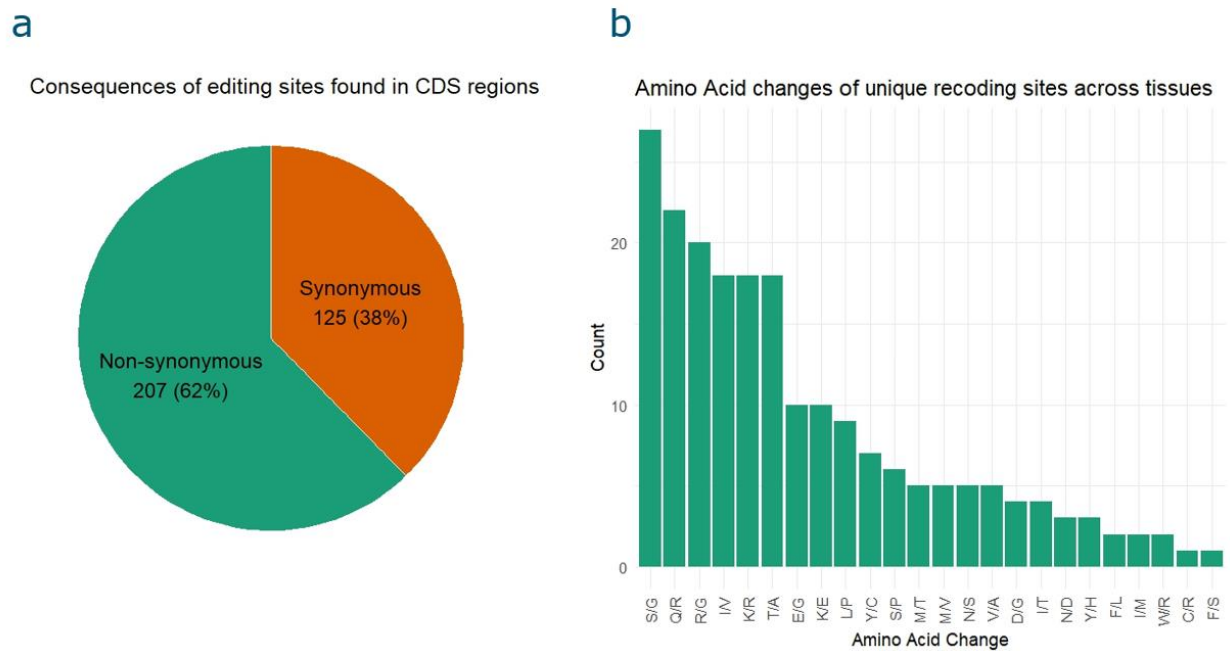


Fig 9 a) Number of non-synonymous and synonymous editing sites located in CDS regions across tissues. b) Frequencies of amino acid changes in recoding sites across tissues.

GO enrichment analysis of genes that have at least one editing site in their CDS region resulted in nominally significant enriched terms for action potential, synaptic signalling and RNA processing (Fig.10). RNA editing is known to affect genes that are involved in neurotransmission (Behm & Öhman, 2016), however, RNA editing activity on transcripts that code for RNA processing proteins is an unexpected find and not reported in literature.

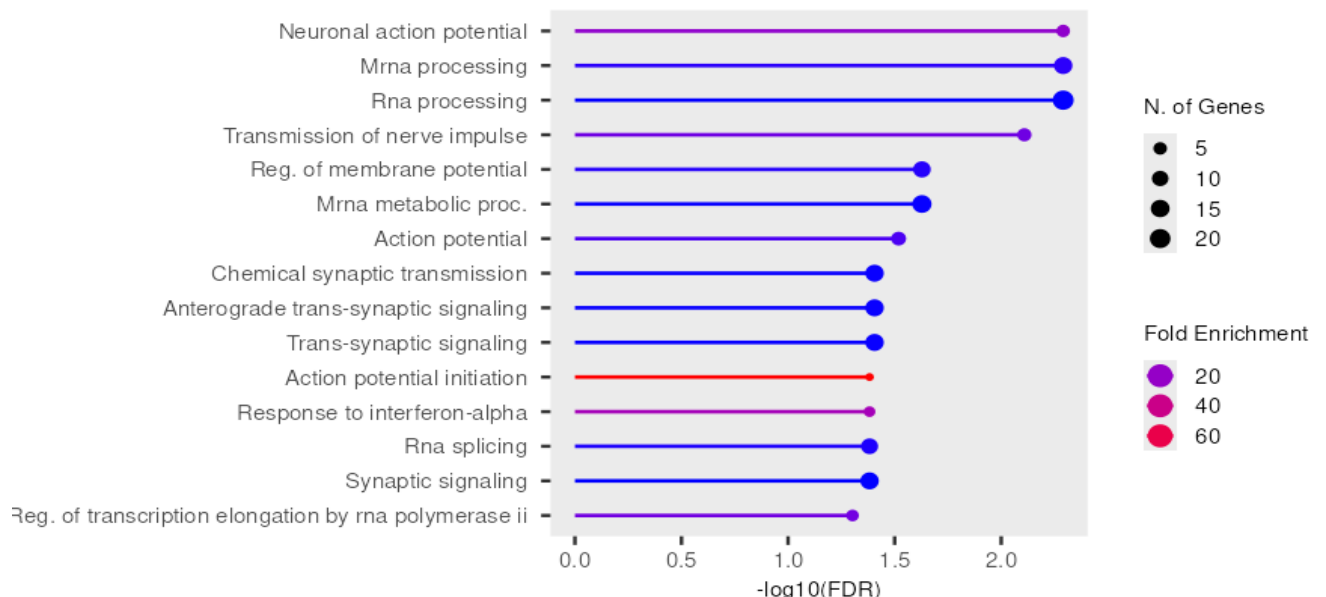


Fig 10. Gene Ontology Enrichment Analysis of genes that have at least one editing site in their CDS region.

Regarding the neurotransmission genes that have editing sites in their CDS region, we detected several editing sites in well-known neuroreceptor genes and

genes regulating neurotransmission. The well-studied *GRIA2* editing site (8:16915172), responsible for the Q/R substitution that is essential for normal functioning of the AMPA glutamate receptor (Lomeli et al., 1994) was captured in all three tissues. A second site in the CDS region of the *GRIA2* was captured in cortex and hypothalamus samples but not in hippocampus samples. Provided that gene expression is high among the three tissue types, this suggests that editing activity can differ between editing sites that lie in the CDS region of the same gene. Furthermore, five editing sites were identified in the *GRIK1* gene, which codes for a subunit of a kainate type glutamate receptor. Three sites were detected in the *KCNQ3* gene and one site in the *KCNA1* gene, both of which code for subunits of different potassium voltage-gated channels and two sites in the calcium voltage-gated channel gene *CACNA1D*. Unlike the two editing sites in the *GRIA2* gene, conclusions about tissue-specific editing of these sites are somewhat risky, since gene expression differs per tissue for these genes and absence of editing in tissues could be due to strict filtering or low coverage and not necessarily due to tissue-specific editing.

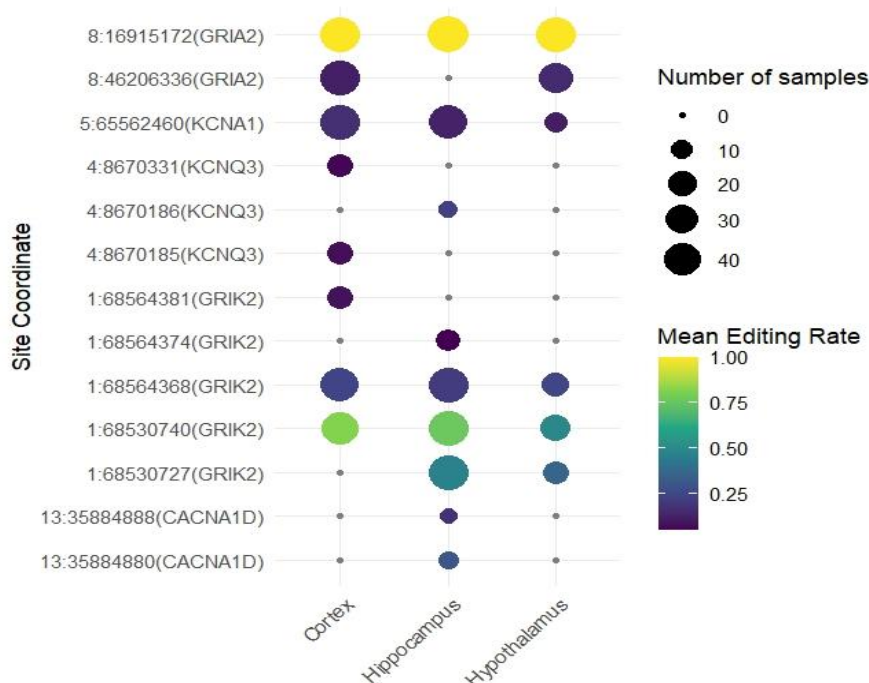


Fig 11. Bubble plot of editing sites located in the CDS region of neuroreceptor genes known to be edited, separate by tissue. On the y axis are the editing site coordinate, as in chromosome:coordinate(gene). On the x axis are the tissue types. Size of each bubble indicates the number of samples a given site is supported by. Colour intensity indicates the mean editing rate of the supporting samples for a given site.

Editing sites in CDS regions are also found in the transcripts of genes that do not code for neuroreceptors but have a regulatory role in neurotransmission. It should be noted, though, that these sites are lowly edited. The genes *C1QL1* and *C1QL3* had each one editing site on their coding sequence and may be implicated in the regulation of the number of excitatory synapses formed in hippocampal neurons. Moreover, an editing site was found in the CDS of *BSN* in cortex and hippocampus, which codes for a protein that regulates pre-synaptic vesicle formation and an editing site in the

CACNG6 CDS region, which codes for a protein that regulates the activity of CACNA1D calcium channels and was edited in all tissue samples.

Highly edited sites were selected as the editing sites with an editing rate of at least 0.65. The majority of highly edited sites lied in 3'UTR and introns (214 and 127 sites, respectively), with only a few sites in 5'UTR and CDS regions. Genes with highly edited sites in CDS involved in neurotransmission were *GRIK2* and *GRIA2*, along with a few non-neurotransmission genes, namely *EEA1*, *RTKN2*, *GBP1*, *PHGDH* and a novel protein-coding gene of unknown function.

3.7 Flanking sequence motif of RNA editing sites

It has previously been shown that there is a particular motif for sequences flanking the editing site (Eggington et al., 2011; Lehmann & Bass, 2000). Investigating the flanking sequences of the editing sites (Fig.12a) revealed a depletion of Guanine in the first position upstream of the site and an enrichment of Guanine in the first position downstream of the site, compared to random control Adenines (Fig.12b) in the genome, in line with previous results.

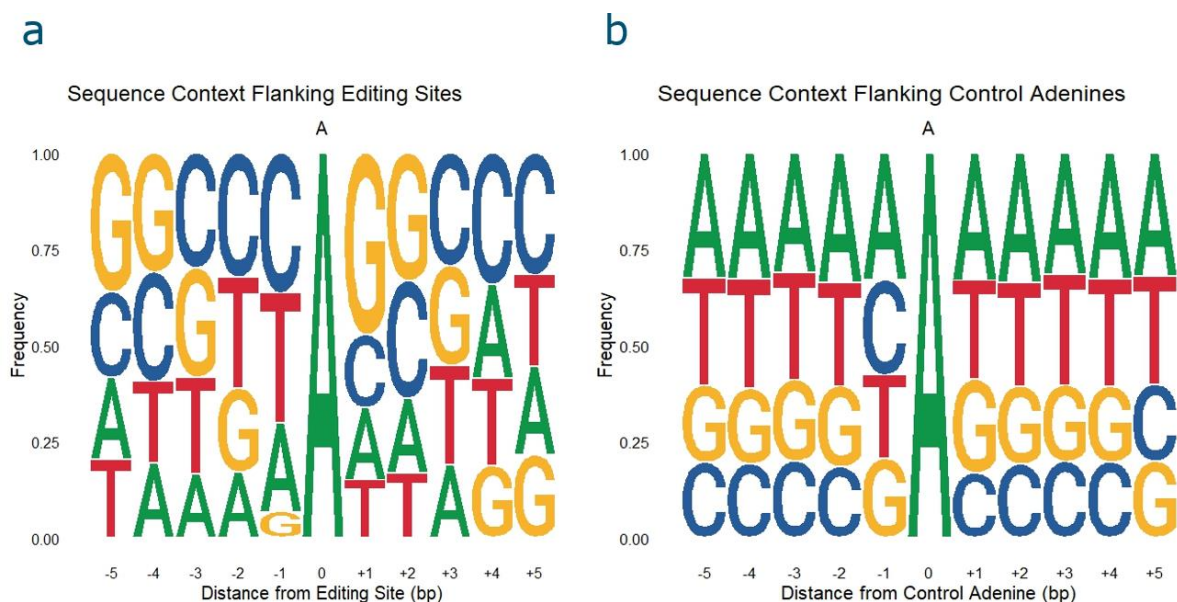


Figure 12. Nucleotide frequency of the first 5 positions upstream and downstream of (a) RNA editing sites and (b) random control Adenines in the genome.

3.8 ADAR expression levels and association with RNA editing activity

To examine the variance in RNA editing that can be explained by the expression of the three ADAR enzymes, we regressed the overall editing levels of samples against the three ADAR enzyme mRNA expression levels per tissue. We failed to find a statistically significant correlation between RNA editing activity and ADAR expression levels in the three brain tissues, except for a weak correlation of ADAR2 expression levels and RNA editing activity in hippocampus samples ($p = 0.03$, $r = 0.175$, Table 2). Note that a few models had marginal p-values which became non-significant after correction for multiple testing.

ADAR	Tissue	Estimate	P-value	Adj. p-value	R squared
ADAR1	Cortex	-0.000156	0.0871	0.157	0.0623
ADAR1	Hippocampus	0.000286	0.0443	0.157	0.085
ADAR1	Hypothalamus	0.000223	0.135	0.203	0.0511
ADAR2	Cortex	-0.000028	0.652	0.734	0.00446
ADAR2	Hippocampus	0.000272	0.00305	0.0275	0.175
ADAR2	Hypothalamus	0.00000241	0.985	0.985	0.00000838
ADAR3	Cortex	-0.000497	0.0779	0.157	0.066
ADAR3	Hippocampus	0.000435	0.199	0.256	0.0356
ADAR3	Hypothalamus	-0.00105	0.0681	0.157	0.0753

Table 2. Parameters for the different models that were constructed to test association of ADAR mRNA expression levels and overall editing level per sample.

3.9 Differentially edited sites between experimental conditions and tissues

The pig population that the RNA-seq dataset was acquired from was subjected to a 2x2 experimental design to compare high versus low allostatic load and enriched versus barren environmental conditions. We therefore investigated differential editing between both experimental factors. The parametric and non-parametric approaches yielded no significant differential editing between conditions for either of the two experimental factors.

We next examined differential editing between tissues. Regarding the cortex and hippocampus comparison, out of the 6,783 tested shared sites supported by at least 12 samples (Fig.14), 144 sites were found to be significantly differentially edited. When hypothalamus editing sites are tested against cortex and hippocampus, out of the 5,004 and 5,410 shared editing sites, respectively, the numbers of significantly

edited sites rise to 801 and 934. Combined with the previous results from PCA and the sample overall editing levels, we conclude that the activity and regulation of RNA editing is very similar between cortex and hippocampus, while RNA editing is more distinct in the hypothalamus compared to the other two tissues.

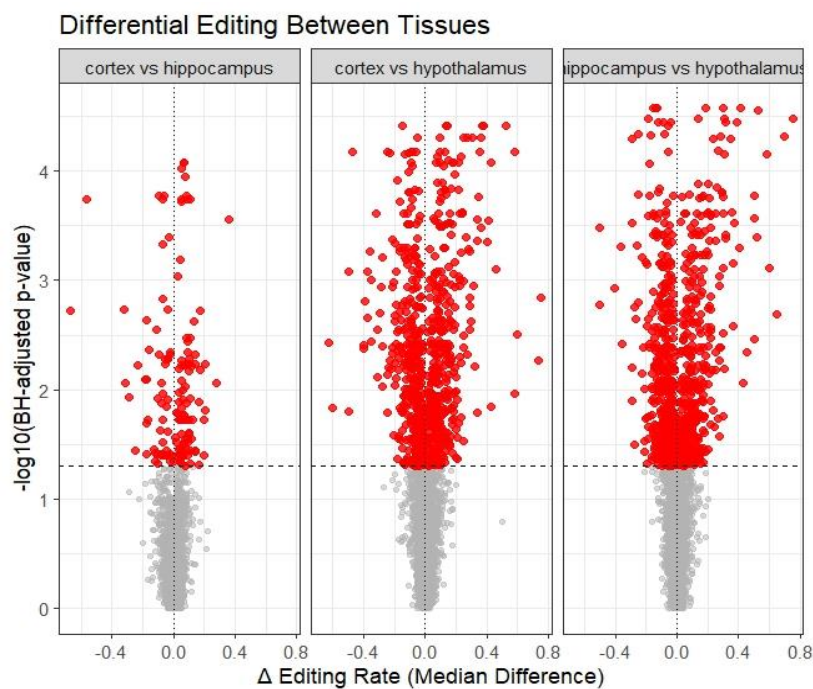


Fig 14. Volcano plots of differential editing sites in the three tissue pairs that were tested. Red dots indicate significant differential editing sites and grey dots indicate non-significant editing sites

Out of the total 1378 unique differentially edited sites among all the comparisons, 732 sites were located in 3'UTR, 389 sites in introns and 30 sites in CDS regions. In the CDS regions, three editing sites are found in *GRK2*, three sites are found in *SON*, two sites in *NOVA1*, which is a splicing factor expressed in the brain (Irimia et al., 2012) and one in *C1QL1*.

4. Discussion

In the present study, we explored the brain RNA editing landscape in a population of guinea pigs subjected to a 2x2 experimental design and compared RNA editing activity across three brain tissues. To our knowledge, this is the first study that attempts to explore RNA editing in a large population of non-human subjects and also the first study to explore RNA editing in the pig brain at the population level. We are also the first that attempt to examine differential editing between three different tissues in the pig brain. We relied on *de novo* identification and strict filtering to detect 15,288 unique canonical A-to-I RNA editing sites across the three tissue types, with most sites shared between at least two tissues. We showed that cortex and hippocampus samples had a similar RNA editing profile, with hypothalamus samples exhibiting a more distinct RNA editing pattern. Hippocampus samples generally had a higher number of editing sites but edited at a lesser intensity compared to cortex and hypothalamus. The majority of RNA editing sites were part of protein-coding genes, with about half of all detected editing sites found in 3'UTR regions, while most editing sites were also located inside the PRE SINE element. A small proportion of editing sites were lying in CDS regions of genes that either code for neuroreceptors, such as *GRIA2* and *GRIK2*, or are known to be involved in neurotransmission. We found a significant association between ADAR2 expression levels and RNA editing activity in hippocampus samples but not in other tissues and ADAR enzymes. We also did not detect significant differential editing between experimental conditions, but rather between brain tissues. Cortex and hippocampus had few differentially edited sites, whereas hypothalamus had many when compared to both cortex and hippocampus.

Accurate detection of RNA editing sites critically depends on the quality of read alignment, as mapping artifacts can easily be mistaken for editing events (Lo Giudice et al., 2020). The splice-aware STAR aligner was used for mapping of the RNA-seq reads to the reference genome with default parameters. Older aligners like BWA (Li & Durbin, 2009) that were not splice-aware failed to map reads that had many mismatches caused by hyper-edited regions. Thus, a pipeline designed to effectively map these hyper-edited RNA reads (Porath et al., 2014) that are unmapped by BWA is often implemented in relative studies. We chose not to apply this pipeline here since STAR can sufficiently tolerate mismatches in the reads during mapping and hyper-edited regions lie primarily in repetitive elements in introns and have both low coverage and low editing rate (Xu et al., 2023). Indeed, by examining a few of the summary output files of STAR, no reads were categorized as unmapped due to too many mismatches. Note that in some cases, the hyper-edited part of a read would get soft-clipped by STAR due to a high number of mismatches, however after alignment of the read the soft-clipped part is omitted from downstream analysis, leading to a potential loss of meaningful editing signal. Additionally, because the mapping step was already conducted before the start of the study, choosing to run STAR in two-pass mode would be preferred to avoid spurious candidate editing sites caused by misalignment in novel splice junctions (Diroma et al., 2021).

De novo identification of RNA editing sites was implemented with JACUSA2, based on the previous tool JACUSA (Piechotta et al., 2017) by the same group. Both JACUSA2 and its predecessor have been featured in high-quality publications (Bannister et al., 2022; Cuddleston et al., 2022; Di Giorgio et al., 2020; Rodriguez de los Santos et al., 2024). A more popular choice based on number of citations in literature is also REDIttools (Picardi & Pesole, 2013), on which the REDlportal database (Mansi et al., 2020) of RNA editing sites in human is based, as is also RES-Scanner (Wang et al., 2016), which requires both RNA-seq and WGS data to operate, hence its invalidity for this analysis. JACUSA2 was favoured over REDIttools due to its fewer dependencies and easier use, as well as its statistical test that is optimized for RNA-seq only workflows with multiple replicates available (Piechotta et al., 2017), whereas REDIttools lacks in that regard. In the end, however, the p-values of the validity of the identified RNA variants by the tool were not considered during filtering for true RNA editing sites, as is common in literature to not rely on statistical tests for editing site inference. As mentioned previously, *de novo* identification with JACUSA2 was performed in three separate sample batches based on tissue. This was done to provide a larger sample size for the statistical test, however, if one of the samples did not pass the filter for a given candidate site, then that site would be removed entirely from the analysis. In hindsight, it would have been more efficient to instead input JACUSA2 with one BAM file at a time, since this approach is overly conservative.

The de novo identification analysis yielded 15,288 unique RNA editing sites identified in total across samples. Interestingly, 70% of the unique sites in the current study are present in a comprehensive pig study examining the brain-wide RNA editing landscape (Huang et al., 2021) and also 83% of the unique sites are present in the PRESDB database (Wang et al., 2019), generated from an older study examining RNA editing across different pig tissues including brain. Although this indicates that true RNA editing sites were picked up, the total number of identified unique sites is significantly lower compared to the aforementioned studies, where the number of detected sites reached about 680,000 and 500,00 correspondingly, as well as others examining RNA editing in different mammals (Bakhtiarizadeh et al., 2018; O'Neil et al., 2017). Also, other pig studies (Yang et al., 2019; Zeng et al., 2026; Zhang et al., 2022; Zhang et al., 2019) with fewer samples that investigate RNA editing in non-brain tissues, which are known to have less editing activity than brain tissues (Tan et al., 2017; Wang et al., 2019), also reported a higher number of identified editing sites.

Several reasons can explain the lower number of unique editing sites and absence of well-studied edited genes. First, the RNA extraction protocol relied on poly-A selection of mature mRNA molecules which limit identification of sites to primarily 3'UTR regions and therefore intronic regions, which is where the bulk of RNA editing takes place, were underrepresented (Nishikura, 2010). Second, we retained editing sites where the standard filters of minimum coverage, minimum number of edited reads and minimum editing rate were satisfied in at least 5 samples in a single tissue. Although this strict filtering approach may cut back on the total number of identified editing sites, this also allows us to better distinguish true from false RNA editing sites when compared to previous studies in pig (Huang et al., 2021; Yang et al., 2019; Zeng et al., 2026; Zhang et al., 2022) with quite lower sample sizes. The detection of RNA

editing events also depends strongly on sequencing depth. Here, an average of 29 million unique reads per sample provides sufficient depth for gene expression analysis, however it is well below the mean number reported in high-quality comprehensive studies (Huang et al., 2021; Hwang et al., 2016; Tran et al., 2019). Therefore, we assume that many weakly edited sites, which constitute the majority of editing sites (Bazak et al., 2014) and have an editing rate below the threshold of 0.05, are not captured here. Finally, only three tissue types from the brain were sampled, two of which were hippocampus and prefrontal cortex, which are known to be quite similar in expression (Sjöstedt et al., 2020). When compared to studies that rely on only a few animals but sample extensively different brain regions (Huang et al., 2021; O'Neil et al., 2017), one could hypothesize that the constrained set of expressed genes in this dataset does not allow the capture of more editing sites.

We were also surprised to find so many editing sites lying in intronic elements and intergenic regions, given that we enriched for poly-Adenylated mRNA molecules. An intuitive explanation for this could be due to poor annotation in the porcine genome, where unannotated genes could be present and therefore editing sites in unannotated genes are mislabelled as lying in intergenic regions. Poor annotation can also explain the presence of editing sites in introns. By visually inspecting aligned reads with detected RNA editing sites mapped to intronic regions in JBrowse, we discovered that quite a few reads are mapped just downstream of the 3' ends of genes, implying that they in fact map to unannotated 3'UTR regions instead of introns. Another possible reason why there were detected editing sites in intronic regions is due to intron retention (Zhang et al., 2019), as has been noted in another study, however, we did not manage to find evidence of that during visual inspection. Nevertheless, we did expect to find at least some editing sites to intronic regions, since nascent transcription of RNA molecules is widespread in the brain and intronic RNAs associated with neurotransmission are common in brain tissues, at least in human (Ameur et al., 2011).

Only 332 unique sites in CDS regions were detected across the three tissues altogether, which accounted for 2.2% of the total unique editing sites. The low detection levels of CDS editing sites, relevant to sites that lie in repetitive elements, has been a consistent phenomenon in previous systematic RNA editing sites identification analyses (Bazak et al., 2014; Picardi et al., 2016; Ramaswami & Li, 2013). In the future, it would be interesting to apply a pipeline that was developed on human data (Gabay et al., 2022) and focuses on CDS editing site detection to examine if more CDS editing sites could be captured in pig as well.

Several editing sites that are known to be found in the coding sequences of neurotransmission genes were not captured in this analysis. This is partly due to the omission of the sex chromosomes in the analysis, which harbor well-studied editing sites, specifically five sites in the *HTRC2* gene (Larsen et al., 2016a) that codes for a serotonin receptor, a site (Ohlson et al., 2007) in the *GABRA3* gene that codes for a Gamma-Aminobutyric Acid Type A receptor and a site (Pinto et al., 2014) in *GRIA3* that codes one of the subunits of the AMPA receptor. CDS editing sites were also not found for *GRIA4*, *GRIK1* and *GRM4* neuroreceptors. Although the sites in *GRIK1* and *GRM4* are reported to be lowly edited in the pig brain, the editing site in *GRIA4* is

known to be highly edited (Huang et al., 2021). Interestingly, however, we report five editing sites in the CDS region of GRIK1, whereas in the pig brain-wide study only four are detected (Huang et al., 2021).

The number of editing sites and the overall editing levels were used to measure RNA editing activity at the sample level. Although both constitute simple and intuitive metrics, they depend on sample coverage which affects how many editing sites (Lo Giudice et al., 2020; Roth et al., 2019) can be detected. Also, tissue-specific gene expression means that the composition of editing sites will differ between samples that originate from different tissues. These problems limit the efficacy of these metrics for inter-sample comparison of RNA editing activity. Despite the bias of tissue-specific expression being unavoidable, the coverage bias can be mitigated with the Alu Editing Index, which has been developed for RNA editing in human (Roth et al., 2019). Alu is the predominant SINE element in the human genome and Alu elements harbor a large amount of editing sites (Bazak et al., 2014). By counting the edited reads in the known locations of the SINE elements and dividing by the total coverage at the Alu editing sites, the background is highly similar between samples, and a more objective global metric can be acquired. However, the application of this approach is uncommon in pig or other mammals, since only primates harbor Alu elements and the SINE elements that are common in other mammal genomes have different distributions and editing frequencies than human Alu elements (Eisenberg et al., 2005). Lastly, the Recoding Index is a metric similar to the overall editing, but instead of including all detected RNA editing sites, it takes into account only recoding sites, meaning sites that lead to a non-synonymous mutation (Lo Giudice et al., 2020; Silvestris et al., 2019). The recoding index is well-suited to study RNA editing in different brain regions and is also correlated with ADAR2 expression (Silvestris et al., 2019), however, a small amount of detected recoding sites could lead to inaccurate results and was therefore not utilized in the present study.

A positive association between ADAR1 and ADAR2 mRNA expression levels and RNA editing activity in the brain has been shown in human (Cuddleston et al., 2022; Tran et al., 2019), pig (Huang et al., 2021) and macaque (Li et al., 2025), but not in cattle (Bakhtiarizadeh et al., 2018), while ADAR3 mRNA expression has been shown to have a negative (Tan et al., 2017) correlation in mammals or no (Breen et al., 2019) association at all in human. Also, a study in mice and human has shown a positive correlation between ADAR1, ADAR2 mRNA expression levels and RNA editing in multiple tissues and a pig study has found significant correlation for ADAR1 expression for seven different tissues (Zhang et al., 2019), including brain, but no significant correlation for ADAR2. In the present study, we were unable to find significant association between ADAR1 or ADAR3 mRNA levels and RNA editing activity and only found a significant positive correlation between ADAR2 mRNA expression and RNA editing in the hippocampus. Two reasons can account for this: firstly, since our samples were correlated due to each guilt being sampled three times, once per tissue, we were forced to model each tissue separately per ADAR enzyme and apply multiple testing correction, which adjusted the p-value of a few tests to non-significant levels. Furthermore, the previously mentioned studies that have been conducted in non-human species rely only on a few animals, therefore it could be that

at the population level, the differences in mRNA expression levels of the ADAR enzymes cannot capture the variance in RNA editing activity.

Differential editing analysis between tissues resulted in few differentially edited sites between cortex and hippocampus and several hundred between hypothalamus and cortex, as well as hypothalamus and hippocampus. Although this is the first study to examine differential editing of sites in the pig brain, it should be noted that we relied on the Wilcoxon signed-rank test to perform it. This test has been applied in other studies as well (Roth et al., 2018; Srivastava et al., 2017; Tran et al., 2019) due to its simplicity, but it fails to take into account the uncertainty in the read counts that arises from insufficient sequencing depth (Tran et al., 2020). A more recent and elegant approach that considers this is to use hierarchical modelling and beta-binomial models (Tran et al., 2020).

Due to time constraints, a few topics were not further investigated or explored at all in the present study. First, we calculated only canonical A-to-I substitutions out of all twelve possible substitutions during the identification of RNA editing sites, hence missing out on background noise and the False Discovery Rate of the *de novo* identification analysis. It would be interesting in the future to look in which genic regions do hyper-edited regions lie and to what extent they are edited. Also, the clustering tendency of editing sites among genic regions could also have been examined, since it has been shown in pig (Huang et al., 2021) that editing clusters, which are editing sites that lie in close proximity to one another, are more likely to be found in 3'UTR regions or in Long Interspersed Nuclear Elements. Furthermore, predicting the functional effect of RNA editing sites in 3'UTR with regards to miRNA binding is also of interest, since RNA editing can alter the binding affinity of miRNA molecules to mRNA, as well as create novel binding sites for miRNAs (Long et al., 2024).

The present research offers a basic exploration of the brain RNA editing landscape in a pig population and provides a foundation for other analyses. A few studies have provided evidence of allele-specific RNA editing in mouse and human (Park et al., 2017; Zhou et al., 2018), which is not only an under-studied biological phenomenon, but also no allele-specific RNA editing study has been conducted on pig as of yet. Quantitative Trait Loci (QTL) mapping has been utilized extensively to elucidate the regulatory mechanisms of molecular phenotypes, such as gene expression QTL (Westra & Franke, 2014). To determine the influence of genetic variants on RNA editing levels, RNA editing quantitative trait loci (edQTLs) have been applied to human brain (Cuddleston et al., 2022; Rodriguez de los Santos et al., 2024) and multiple pig tissues, including brain (Wang et al., 2019). An edQTL analysis across different pig brain tissues is lacking in literature and could be an interesting next step in the study of pig brain RNA editing. Phenotypic data were also recorded in the pig population that this study was based on. Linking pig phenotypic traits with RNA editing levels of sites could provide an interesting insight into the biological implications of RNA editing. Furthermore, the brain is known to have a rich cellular diversity and brain subregions can have a highly different cellular composition from one another (Sjöstedt et al., 2020). In order to better understand the differences in RNA editing across pig brain tissues, *in vivo* single-nuclei RNA-seq or cell-type-specific brain nuclei

populations can be used to investigate which brain cell types are edited and to what extent, as has been done in human studies (Cuddleston et al., 2022; Cuddleston et al., 2022).

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